

Name: _____

Date: _____

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MyDesign Portfolio

Element H

Priorities

1. Opening the gate full extension
2. Opening the gate within 3 seconds
3. Make sure it doesn't break in 10 uses
4. 3 random students understand how to use it with no help
5. Clear indicator of gate being open or closed

requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
The gate must open to the full extension possible	1) Close blast gate all the way 2) Use controller and open the blast gate 3) Measure how much of the door is visible in the hole	4 trials of closing the gate manually, then opening it with the controller	The door will either open all the way, only open partly, or not move at all	Fail: More than 1 inch of the door is visible Pass: Less than 1 inch of the door is visible Goal: Have $\frac{3}{4}$ of an inch of the door still visible
The gate must open quickly	1) Close blast gate manually 2) Use controller to open the gate	6 trials of closing the gate manually, then opening it with the controller	The door will take between 0.5 to 3 seconds, and could also not move at all	Fail: The door takes more than one full second to open Pass: Less than 1 second to open Goal: take 0.5 seconds to close
Make sure it doesn't break	1) Open and close the blast gate	20 trials of closing then opening the gate	The door will break completely,	Fail: The mechanism breaks after less

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		with the controller	break partially, or not at all	than 10 trials Pass: The mechanism breaks after more than 10 trials Goal: The mechanism doesn't break
Students should be able to use it without help	1) Provide instruction of how to operate the door to a student 2) Allow the student to attempt to follow directions	5 different students are given the chance to operate the door	A student may need another demonstration, have some idea of what to do, or confident in their ability to operate the doors	Fail: Less than 3 students know how to operate the door Pass: At least three students know how to operate the door Goal: All five students know how to operate the door